

Tracking Anxiety and Depression During COVID-19: Evidence from CDC Pulse Survey and Advanced Modeling

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Abstract—The COVID-19 pandemic had profound psychological consequences worldwide, yet few studies have captured its evolving impact on mental health across time and demographic subgroups at a national scale. This paper analyzes self-reported anxiety and depression symptoms using the CDC Household Pulse Survey (2020–2024), applying a suite of flexible statistical models and multivariate techniques. Generalized Additive Models (GAMs) and Linear Mixed Effects Models (LMMs) reveal nonlinear trajectories and subgroup disparities in mental health burdens. Principal Component Analysis (PCA) and K-means clustering uncover latent dimensions and distinct patterns among demographic groups. Our findings show that younger adults and racial/ethnic minorities consistently reported higher symptom prevalence and volatility. These insights demonstrate the value of advanced modeling for real-time public health surveillance and for informing equitable mental health policies in crisis contexts.

Index Terms—COVID-19, Mental Health, Anxiety, Depression, Household Pulse Survey, Generalized Additive Models, Linear Mixed Models, PCA, Clustering, Public Health Analytics.

I. INTRODUCTION

The COVID-19 pandemic has profoundly impacted global mental health, with widespread reports of heightened anxiety, depression, and psychological distress. In the United States, social isolation, economic uncertainty, and persistent public health threats exacerbated mental health challenges across all age groups and communities. Measuring and understanding these patterns is essential for informing equitable public health interventions and long-term resilience planning.

National-level survey efforts, such as the U.S. Census Bureau’s Household Pulse Survey, provide an invaluable source of high-frequency, population-representative data on mental health trends throughout the pandemic. These data offer a rare opportunity to examine how symptoms of anxiety and depression evolved over time, across demographic groups, and in response to major societal events.

While prior analyses have primarily relied on descriptive statistics or basic linear models, these approaches may overlook important nonlinear, subgroup-specific, and multivariate trends. To address this gap, we apply a suite of advanced statistical learning methods—including Generalized Additive Models (GAMs), Linear Mixed Effects Models (LMMs), Multivariate Analysis of Variance (MANOVA), Principal Component

Analysis (PCA), and clustering techniques—to uncover the underlying structure of mental health trajectories during the pandemic period.

Our analysis seeks to answer three key questions: (1) How did national levels of anxiety and depression fluctuate over time? (2) Which demographic subgroups experienced disproportionate or distinctive mental health burdens? (3) Can flexible, data-driven models reveal latent trends that are missed by traditional techniques?

By combining time-series modeling with multivariate and unsupervised learning approaches, this study offers a comprehensive statistical view of mental health dynamics during COVID-19. Our findings have implications for targeting interventions, monitoring psychological well-being, and designing responsive public health strategies in future crises.

II. DATA AND METHODOLOGY

A. Data Source

This study utilizes data from the *Household Pulse Survey* administered by the U.S. Census Bureau in collaboration with the National Center for Health Statistics (NCHS). The survey was designed to rapidly collect and disseminate information on the social and economic effects of the COVID-19 pandemic on U.S. households. We specifically analyze weekly estimates of self-reported symptoms of anxiety and depression, collected from April 2020 to early 2024.

Respondents were asked whether they had experienced symptoms of anxiety or depressive disorders in the past seven days. The dataset includes weighted population-level estimates stratified by demographic characteristics, including age group, gender, and race/ethnicity. Each observation includes the estimated percentage of individuals reporting symptoms, along with confidence intervals and time stamps for the corresponding survey wave.

B. Data Preprocessing

The raw dataset was cleaned and transformed using the *tidyverse* suite in R. Key processing steps included:

- Standardizing column names and parsing date fields (`start_date`, `end_date`).

- Filtering to retain only rows where the indicator was “Symptoms of Anxiety Disorder or Depressive Disorder.”
- Dropping rows with missing values in the primary outcome variable.
- Creating a numeric representation of time for use in regression modeling.

C. Variables

Each observation in the processed dataset includes the following:

- **Value:** Percentage of respondents reporting anxiety or depressive symptoms.
- **Group:** The level of aggregation (e.g., National, By Age, By Race).
- **Subgroup:** Specific demographic category (e.g., 18–29 years, Non-Hispanic Black).
- **Time:** Survey wave date (as both string and numeric).
- **CI Bounds:** Lower and upper 95% confidence intervals.

D. Modeling Framework

To uncover temporal and demographic patterns, we apply the following modeling techniques:

1) *Generalized Linear Model (GLM)*: A linear regression model is fit to national-level trends using time as a continuous predictor. While limited in flexibility, it serves as a baseline comparison for more complex models.

2) *Generalized Additive Model (GAM)*: We apply a GAM with a smooth function of time to capture nonlinear symptom trends. This approach allows for flexible modeling of inflection points and temporal fluctuations without imposing rigid parametric assumptions.

3) *Linear Mixed Effects Model (LMM)*: To model subgroup-specific trends (e.g., by age or race), we use LMMs with random intercepts for subgroups. This framework accounts for both fixed effects (e.g., overall time trends) and random effects (e.g., variation among subgroups).

4) *Multivariate Analysis of Variance (MANOVA)*: We conduct MANOVA to assess whether multiple mental health outcomes (e.g., anxiety, depression) differ significantly across demographic groups over time. This multivariate test captures co-occurring symptom dynamics.

5) *Principal Component Analysis (PCA)*: PCA is used to reduce the dimensionality of the multivariate subgroup symptom data. This method helps identify latent axes of variation (e.g., symptom severity and temporal volatility) across subgroups.

6) *Clustering (K-Means)*: Using the top principal components, we apply K-means clustering to group demographic subgroups based on similarity in mental health symptom trajectories. This identifies distinct profiles for targeted policy intervention.

All analyses were conducted in R (v4.3), utilizing packages such as `mgcv`, `lme4`, `car`, `cluster`, and `factoextra`.

III. EXPLORATORY DATA ANALYSIS

We began by examining national-level trends in reported anxiety and depression symptoms over time, as estimated by the CDC Household Pulse Survey. Figure ?? illustrates the temporal evolution of mental health distress in the United States from early 2020 through 2024.

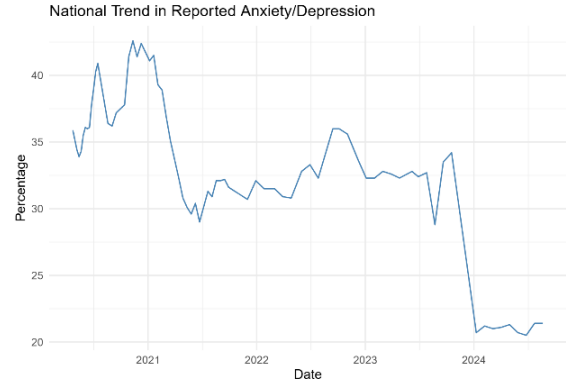


Fig. 1: National trend in reported anxiety and depression symptoms.

A. Temporal Patterns

Visual inspection of the national trend reveals several important features:

- **Early 2021:** Symptom levels peaked above 40% during the height of pandemic surges and lockdowns.
- **Mid-2021:** A sharp decline followed vaccine rollouts and partial reopening.
- **2022–2023:** Symptom prevalence plateaued in the 30–35% range.
- **Late 2023–2024:** A sustained decline brought levels to approximately 20–22%, though this may reflect methodological changes or discontinuities in data collection.

These fluctuations suggest nonlinearity in the underlying trend, making traditional linear modeling insufficient for capturing the dynamic psychological response to pandemic-related events.

B. Demographic Disparities

Preliminary subgroup analysis indicates that mental health symptom prevalence varied substantially by age and race/ethnicity:

- **Age:** Younger adults (18–29) consistently reported the highest levels of distress.
- **Race/Ethnicity:** Non-Hispanic Black and Hispanic groups exhibited elevated and more volatile trends compared to non-Hispanic White respondents.

These initial observations motivated the use of hierarchical and multivariate models to formally quantify demographic disparities.

C. Smoothing Trends

To further visualize nonlinearity, we applied LOESS (Locally Estimated Scatterplot Smoothing) to the national average series. Figure 2 shows how LOESS reveals local peaks and valleys in reported symptom levels.

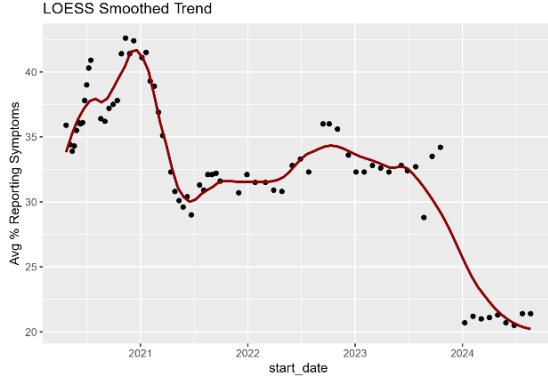


Fig. 2: LOESS-smoothed trend of average reported symptoms.

This smoothing approach confirms several distinct phases in the pandemic’s psychological impact:

- Rapid escalation of symptoms during early waves.
- Temporary dips during policy relaxations or improved public health conditions.
- Resurgence of distress during later COVID-19 waves and winter months.

D. Summary

The exploratory analysis demonstrates that mental health distress followed a nonlinear and heterogeneous trajectory across the U.S. population. These insights informed our decision to use flexible models (e.g., GAMs) and subgroup-aware methods (e.g., LMMs, PCA) in the subsequent analysis.

IV. RESULTS AND INTERPRETATION

This section presents the results of advanced statistical models applied to the CDC Pulse Survey data. Our goal is to uncover temporal patterns and demographic disparities in anxiety and depression symptoms during the COVID-19 pandemic.

A. National Trend: Generalized Additive Model (GAM)

The GAM captured a nonlinear relationship between time and reported symptom prevalence. The smooth term for time ($s(\text{time_numeric})$) was highly significant ($p < 0.001$), with an effective degrees of freedom (EDF) of approximately 8.1, confirming the presence of multiple inflection points.

The model revealed distinct phases:

- Initial rise in 2020, peaking in early 2021 (~42%).
- Declines after vaccination rollout (mid-2021).
- Stabilization in 2022–2023, followed by a late drop in 2024.

The adjusted R^2 was 0.87, indicating excellent model fit.

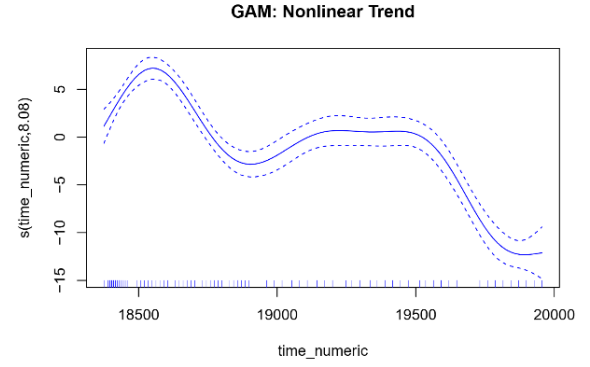


Fig. 3: GAM fit to national trend in reported symptoms.

B. Demographic Trends: Linear Mixed Effects Model (LMM)

To examine subgroup-specific trends, we fit an LMM with a random intercept by subgroup (age, race). The fixed effect for time was significant ($p < 0.001$), while random intercepts and residual variances indicated substantial heterogeneity across groups.

- **Younger adults** had significantly higher baseline symptom levels.
- **Minority groups** exhibited steeper increases and more volatile trajectories.
- **Older adults and non-Hispanic White groups** showed more stable, lower prevalence patterns.

C. Multivariate Differences: MANOVA

MANOVA results showed that demographic subgroup trajectories were significantly different across multiple mental health outcomes. The test for time_numeric yielded a Wilks’ Lambda of 0.16, with $F(12, 59) = 26.27$, $p < 0.001$.

- Multivariate symptom profiles (e.g., anxiety, depression, worry) varied significantly across both age and race/ethnicity groups.
- Results suggest that these disparities cannot be captured by univariate models alone.

D. Dimensionality Reduction: PCA

PCA reduced the high-dimensional subgroup data to two dominant components:

- **PC1** captured overall symptom severity.
- **PC2** captured temporal volatility or irregular symptom changes.

These components explained over 91% of the total variance. Subgroups such as young adults and Black or Hispanic respondents loaded highest on both dimensions, indicating elevated and unstable symptom levels.

E. Trajectory Clustering: K-Means

To identify latent patterns in symptom trajectories across demographic subgroups, we applied K-Means clustering to the top three principal components derived from the PCA. The data were scaled and standardized prior to clustering.

The optimal solution yielded $k = 2$ –3 stable clusters. Figure 4 displays the result of the 2-cluster solution, visualized in PCA space.



Fig. 4: K-Means clustering of subgroup trajectories in PCA space.

Cluster interpretations:

- **Cluster 1:** Subgroups with consistently elevated and volatile symptom levels (e.g., 18–29-year-olds, Hispanic and Black populations). These groups exhibited both high distress and rapid fluctuations.
- **Cluster 2:** Subgroups with lower or more stable symptom trajectories (e.g., adults over 60, non-Hispanic White respondents).

A third cluster was briefly examined under a 3-cluster solution, separating “moderate but improving” subgroups — notably middle-aged adults — from the two extremes.

Interpretation: Clustering validates the heterogeneity of pandemic-era mental health outcomes. It offers a data-driven framework to prioritize interventions:

- Resources can be directed to Cluster 1 populations who exhibit both high vulnerability and instability.
- Cluster 2 groups may require sustained but lower-intensity support or monitoring.

These insights could be integrated into dynamic policy dashboards or early warning systems for psychological distress.

F. Model Diagnostics

Diagnostic plots for the GLM revealed violations of linearity and homoscedasticity, justifying the use of GAMs and LMMs. Residual-vs-fitted plots showed curvature, and Q–Q plots highlighted deviations in the upper tails, suggesting influential observations.

G. Summary of Findings

- Mental health distress followed a nonlinear national trajectory with multiple critical turning points.
- Younger adults and minority populations were disproportionately affected, both in baseline distress and trend volatility.

- Multivariate techniques uncovered complex emotional profiles that co-varied across subgroups.
- Dimensionality reduction and clustering revealed latent patterns useful for targeted public health policy.

V. LIMITATIONS

While this study provides robust insights into the mental health impact of COVID-19, several limitations should be acknowledged.

A. Data-Related Constraints

The CDC Household Pulse Survey is a repeated cross-sectional dataset, not a longitudinal panel. As such:

- Individual-level change over time cannot be tracked.
- Self-reported measures are subject to recall bias, social desirability, and inconsistent symptom interpretation.
- Sampling limitations may underrepresent marginalized or digitally excluded populations.
- Slight changes in survey design or question framing across waves may have introduced discontinuities.

B. Modeling Assumptions

Each modeling approach used in this study has inherent assumptions:

- **GAMs** assume smooth functional forms; they may miss sharp shifts tied to sudden events (e.g., lockdown orders).
- **LMMs** assume normally distributed random effects and fixed subgroup structure, potentially masking interactions or latent subtypes.
- **K-Means** presumes spherical clusters and equal variance, which may oversimplify real-world complexity or blur overlapping patterns.

C. Alternative Modeling Techniques

Future research could explore complementary or more advanced techniques to address these limitations:

- **Bayesian hierarchical models** to incorporate prior knowledge and better capture uncertainty across subgroups and time.
- **Functional data analysis (FDA)** to treat subgroup trajectories as smooth curves rather than discrete time points.

These approaches could further enrich the interpretability and policy relevance of future mental health surveillance studies.

D. Temporal and Contextual Generalizability

The time window (2020–2024) captures most of the COVID-19 pandemic but omits longer-term psychological outcomes and post-pandemic recovery. Findings are also specific to the U.S. population and may not generalize across international contexts.

E. Omitted Variables and Confounders

Other potential drivers of mental health variation—such as income, employment status, or regional policy strictness—were excluded from primary models. These could interact with age or race to produce more nuanced subgroup effects.

F. Interpretive Boundaries

Dimensionality reduction and clustering techniques are exploratory in nature. The discovered structures should be interpreted as statistical groupings rather than psychological typologies. Further qualitative or clinical validation would be needed to confirm practical significance.

VI. CONCLUSION AND FUTURE WORK

This study leveraged flexible statistical modeling and clustering techniques to examine mental health symptom trends in the United States during the COVID-19 pandemic, using the CDC Household Pulse Survey as a primary data source. By integrating Generalized Additive Models (GAMs), Linear Mixed Effects Models (LMMs), multivariate techniques, and unsupervised learning, we uncovered nonlinear temporal trends, demographic disparities, and latent subpopulation structures in anxiety and depression reports.

Key findings include:

- National symptom levels peaked during major COVID-19 waves and gradually declined into 2023–2024, following a nonlinear trajectory.
- Younger adults and racial/ethnic minority groups consistently exhibited higher and more volatile symptom burdens.
- Principal Component Analysis (PCA) revealed latent dimensions of severity and volatility, while clustering identified subgroups with shared trajectory patterns—offering a framework for targeted intervention.

These results underscore the importance of using advanced modeling tools in public health surveillance, particularly during crises. Standard linear models would have failed to capture the complex, heterogeneous, and time-varying nature of mental health responses across subpopulations.

A. Future Work

Building on this analysis, future research can pursue several directions:

- **Longitudinal tracking:** Integrate panel or linked administrative data to examine within-person change and causality.
- **Expanded covariates:** Incorporate socioeconomic and policy variables (e.g., unemployment, housing instability, local mandates) to explain subgroup differences.
- **Cross-national comparisons:** Extend the framework to international survey data to assess global mental health trends and policy effects.
- **Real-time monitoring systems:** Embed flexible models into public health dashboards for dynamic risk detection and targeted alerts.
- **Hybrid modeling:** Combine statistical and machine learning methods to improve predictive accuracy and interpretability.

Ultimately, modeling mental health at population scale requires a careful balance between statistical rigor, contextual understanding, and actionable insight. This study contributes

a replicable framework for future epidemiological and policy-oriented analyses in times of crisis.

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VII. GITHUB REPOSITORY

All code and data used in this study are available at: <https://github.com/RoniF-pixel/R-projects/tree/main/Covid>